6/8/2020

Autophagosome / Lysosome Modeling

$Y(x)$ = spatial density of lysosomes

$P(x)$ = " " of autophagosomes

$A(x)$ = for an autophagosome at position x, what is its acidity

- for now, assume no lysosome reversals. Assume constant fusion rate k_f.

$$\frac{dP}{dt} = v_p \frac{\partial P}{\partial x}$$ (v_p: assumed positive)

$$\frac{dY}{dt} = -v_y \frac{\partial Y}{\partial x} - k_f Y (P \ell)$$ (ℓ: interaction length; $P\ell$: # of phagosomes that can fuse at this position)

change in acidity upon fusion: α = lysosome acidity, constant

$$A_{new} = \frac{A_{old} V_P + \alpha V_L}{V_P + V_L}$$

$$\Delta A = \frac{A_{old} V_P + \alpha V_L}{V_P + V_L} - A_{old} = \frac{\alpha V_L - A_{old} V_L}{V_P + V_L} = (\alpha - A) \frac{V_L}{V_P + V_L}$$

$$\frac{dA}{dt} = v_p \frac{\partial A}{\partial x} - k_f \underbrace{\tilde{Y} \ell}_{\text{lysosomes in fusion range}} (\alpha - A) \frac{V_L}{V_L + V_P}$$

Boundary conditions:

autophagosome flux: $v_p P(x=L) = k_{pp}$ (k_{pp}: autophagosome production rate)

$v_y Y(x=0) = k_{py}$ ← lysosome production rate

$A(x=L) = 0$ ← autophagosomes start deacidified

steady state:

$$\frac{dP}{dt}=0 \Rightarrow \frac{\partial P}{\partial x}=0 \Rightarrow \text{constant phage density} \qquad \boxed{P(x)=\frac{k_{pp}}{v_p}}$$

$$\frac{dY}{dt} = -v_y\frac{\partial Y}{\partial x} - k_f Y \frac{k_{pp}}{v_p}\ell = 0$$

$$\frac{\partial Y}{\partial x} = -\underbrace{\frac{k_{pp}k_f\ell}{v_p v_y}}_{\beta,\ \text{units of } \ell^{-1}} Y \Rightarrow Y(x) = C_1 e^{-\beta x}$$

$$Y(0) = C_1 = \frac{k_{py}}{v_y}$$

$$\boxed{Y(x) = \frac{k_{py}}{v_y} e^{-\left(\frac{k_{pp}k_f\ell}{v_p v_y}\right)x}}$$

$$\frac{dA}{dt} = v_p\frac{\partial A}{\partial x} + \frac{k_f \ell V_L}{V_L+V_p}\frac{k_{py}}{v_y}e^{-\beta x}(\alpha - A) = 0$$

$$\frac{\partial A}{\partial x} = -\underbrace{\frac{k_f k_{py}\ell}{v_y v_p}\frac{V_L}{V_L+V_p}}_{\gamma} e^{-\beta x}(\alpha - A)$$

$$\boxed{A(x) = \alpha + \exp\left[\frac{e^{-\beta x}\gamma}{\beta}\right] C_1} \qquad \alpha - A = -C_1 \exp[\ \ ]$$

$$A'(x) = \exp\left[\frac{-e^{-\beta x}\gamma}{\beta}\right]\left(\frac{-\gamma}{\beta}\cdot -\beta\right) C_1 e^{-\beta x} = +C_1\gamma e^{-\beta x}\exp[\ \ ] = -\gamma e^{-\beta x}(\alpha - A) \ \checkmark$$

boundary:

$$A(L) = \alpha + \exp\left[\frac{-e^{-\beta L}\gamma}{\beta}\right] C_1 = 0$$

$$C_1 = -\alpha \exp\left[+\frac{\gamma}{\beta}e^{-\beta L}\right]$$

$$A(x) = \alpha\left\{1 - \exp\left[\frac{\gamma}{\beta}\left(e^{-\beta L} - e^{-\beta x}\right)\right]\right\}$$

Parameters: α acidity level per lysosome

units of per length:

- β: something about how many fusion-events occur, how far lysosome goes before fusion $\sim K_{pp}$
- γ: fusion events per phagosome? $\sim K_{py}$